ISSUE NO. 01

DATA PORTFOLIO · MANAUS, BRAZIL · OCTOBER 2026

# Data.

*from the chemistry bench to the algorithm*

COVER STORY

## The algorithm linking the Amazon to 384 universities worldwide

Sentence-BERT + OpenAlex across 58,084 research connections




MACHINE LEARNING

Chemistry meets XAI: SVR, SHAP and DFT on one screen

DATA ENGINEERING

Amazonas Fire Department: from shift PDFs to a map dashboard

INSTITUTIONAL BI

A university's internationalization in 6 public dashboards

**William Pinheiro**

PHD IN MATERIALS CHEMISTRY · AI-ASSISTED DATA ANALYSIS

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PROFILE

# A chemist who learned to make data speak

*From the catalysis lab to a university research office: real projects, real data, published and in use.*

William Pinheiro holds a PhD in Chemistry from the Federal University of Amazonas (UFAM), including a sandwich doctorate at the University of Pisa, Italy. He spent years generating data at the bench — XRD, FTIR, HPLC, GC-MS, ICP-OES. That routine of spreadsheets, calibration curves and model fitting sparked his interest in the other half of the job: cleaning, modeling, explaining and communicating.

He is now a postdoctoral researcher at the Office of Research and Graduate Studies of Amazonas State University (PROPESP/UEA), where he builds the dashboards and data pipelines behind the university's internationalization — from scraping and APIs to semantic matching with language models. At the Amazon Bio-business Center (CBA), he developed semi-automated Power BI reports and Python data flows.

His specialty is chemistry; data analysis is the tool. The systems in this issue were built with **generative AI assistance** (Claude Code) for writing code: the questions, the choice of methods, the validation of results and the chemical interpretation are his own.

The common thread is the same as in experimental chemistry: **methodological rigor and honesty about limitations**. Small datasets are treated as small, correlation is not sold as causation, and every dashboard states what the data *cannot* support.

*Every dashboard starts with a decision-maker's question — and is done when that person can answer it on their own.*

**35**
projects · 17 with interactive dashboards, 10 online

**58k**
research connections generated with NLP

**288**
incidents extracted from operational PDFs

**7**
journal articles · 184 citations

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15 **Full index: 35 projects**
everything built, by area

17 **Insert: résumé and Lattes CV**
experience, publications, projects





**William Pinheiro**
PHD IN CHEMISTRY · MANAUS, BRAZIL


**How to read:** every image is a screenshot of the real system (interfaces in Portuguese). Public dashboards have QR codes on the back cover — open them and explore.




COVER STORY · APPLIED NLP

# The algorithm that finds partners for the Amazon

*Turning research lines into vectors to discover, among millions of publications, who in the world studies the same topics as UEA's graduate programs.*

Before every international mission, UEA's research office faced a hard question: *who is worth talking to?* The answer came from manual searches, personal contacts and intuition. **FAAP — Find a Academic Partner** replaced that process with a reproducible pipeline.

The starting point is the official research lines of each graduate program. On the other side, the **OpenAlex** API returns researchers and publications from the target universities. Texts are translated from Portuguese to English, stripped of stopwords and lemmatized with **spaCy**.

The leap in quality came from replacing exact keyword counting with **semantic similarity**: a **Sentence-BERT** model (all-MiniLM-L6-v2) embeds each research line and each publication title, and cosine similarity ranks the candidates. Synonyms and equivalent technical terms are now recognized, and batch ranking processes an entire country in one run.

The same model classifies UEA's 166 research lines into up to three UN **Sustainable Development Goals** — making it possible to cross foreign universities and SDGs in a single matrix.

**29**
graduate programs

**323**
UEA faculty members

**384**
foreign institutions

**58,084**
ranked connections

**10**
countries, from Ghana to Singapore and the US

## THE PIPELINE

**01 · ACQUISITION**
OpenAlex API with caching and daily quota control; ORCID for real publications

→

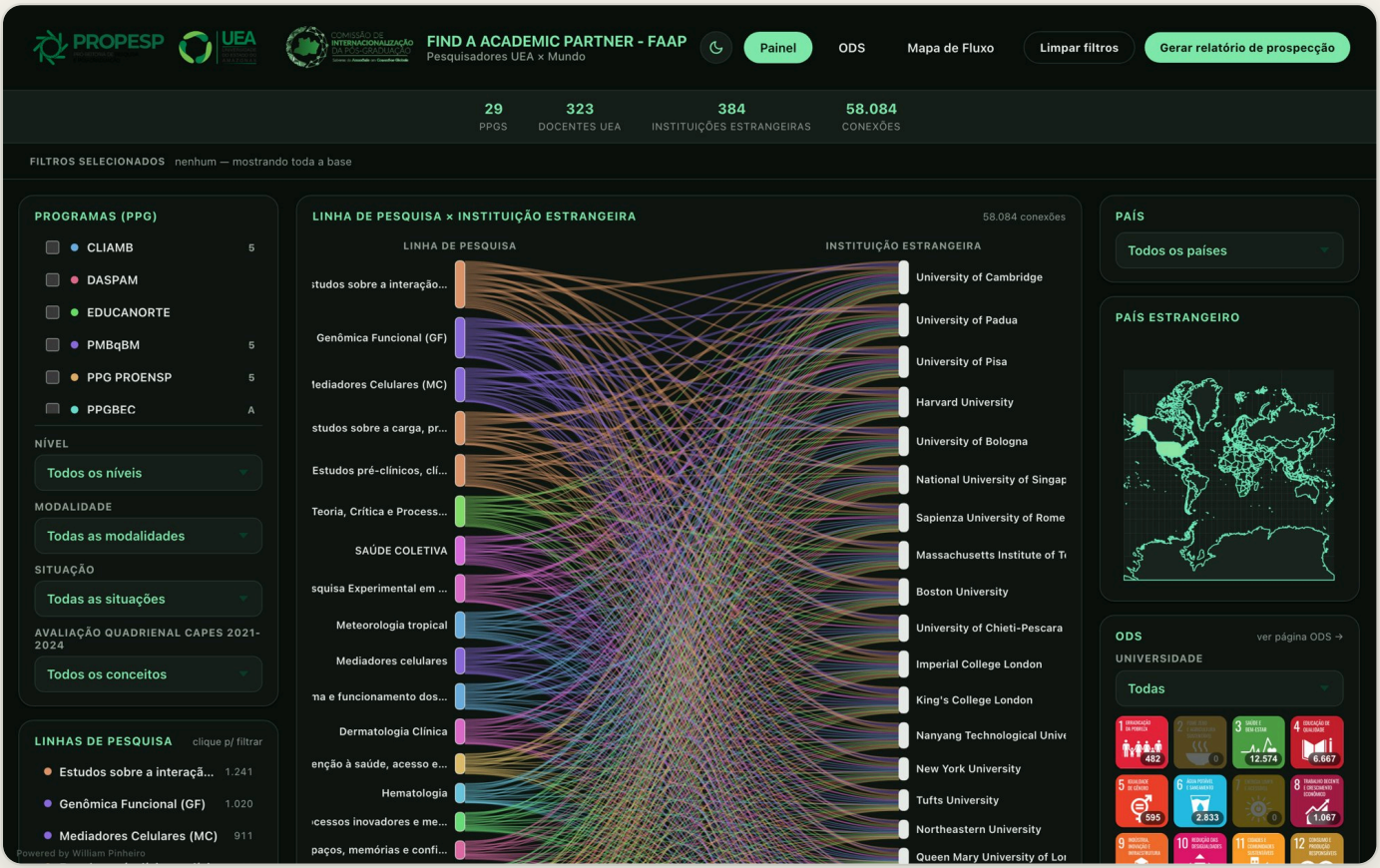
**02 · STANDARDIZATION**
PT→EN translation, stopwords and lemmatization (spaCy)

→

**03 · MATCHING**
Sentence-BERT embeddings + cosine, batch re-ranking per country

→

**04 · PUBLISHING**
Static JSON → dashboard on GitHub Pages, no server




**Main dashboard.** Cascading filter program → research line, Sankey of research line × foreign institution, country map and SDG panel. Every faculty member and researcher opens a profile with real publications.




**2030 Agenda.** Research lines classified into SDGs by semantic similarity, plus a university × SDG matrix.

### WHY IT MATTERS

- It underpins the **prospecting reports** for UEA's international missions (United States and United Kingdom).
- Generic architecture: adding a country is a configuration entry, not new code.
- Researcher "seeds" fill OpenAlex gaps without biasing the ranking.
- The same method was replicated for **institutions already under agreement**.

`Python` `Sentence-BERT` `spaCy` `OpenAlex API` `ORCID` `SQLite` `D3.js` `GitHub Pages`

EXPERIMENTAL DATA SCIENCE

# When chemistry calls for machine learning

*A study on dye removal with Amazonian plant fibers became a modeling lab: explainable ML, response surfaces, nonlinear fitting and quantum chemistry — in a single dashboard.*

Palm and pod fibers — untreated, acid-treated or base-treated — were tested as adsorbents for methylene blue dye. The data lived in a lab spreadsheet on Google Drive, so the first task was engineering: an **ETL** that pulls the latest version through the API, validates the structure of each tab and only then replaces the **SQLite** database. If the spreadsheet layout changes, the process fails loudly instead of silently writing bad data.

Validation exposed formula errors in the original adsorption capacity ($Q_e$) column; values were recomputed from the standard equation.

On the 49 experimental runs, three algorithms — **Random Forest, XGBoost and SVR** — were tuned with `GridSearchCV` and 5-fold cross-validation. SVR performed best on the test set. **SHAP** values show adsorbent mass as the most influential variable, consistent with the physics of the process.

The dashboard also includes a simulator: users move the parameters and a Flask backend returns predictions in real time.

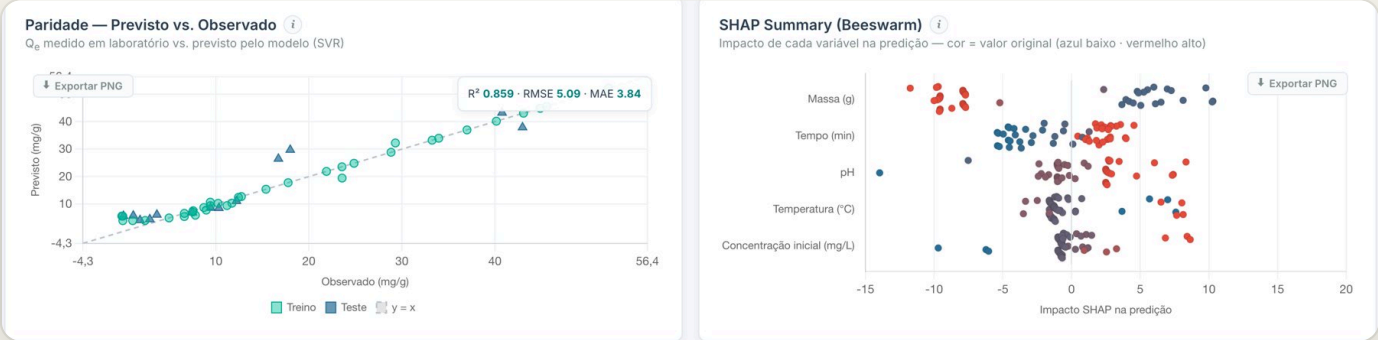
**0.859**
test $R^2$ (SVR, $Q_e$) · n test = 13

**5.09**
test RMSE, in mg/g

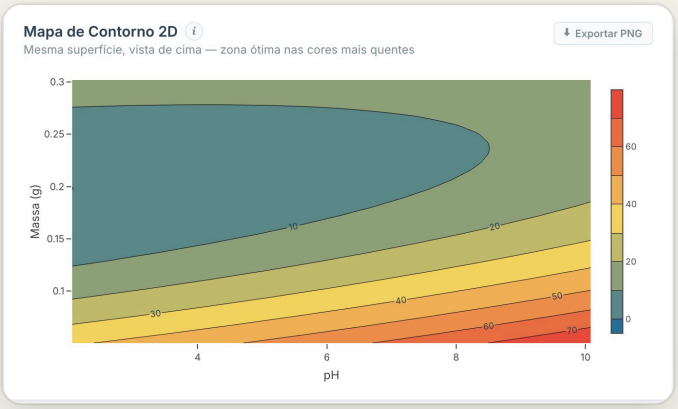
**49**
real runs · 5 process variables

METHOD NOTE

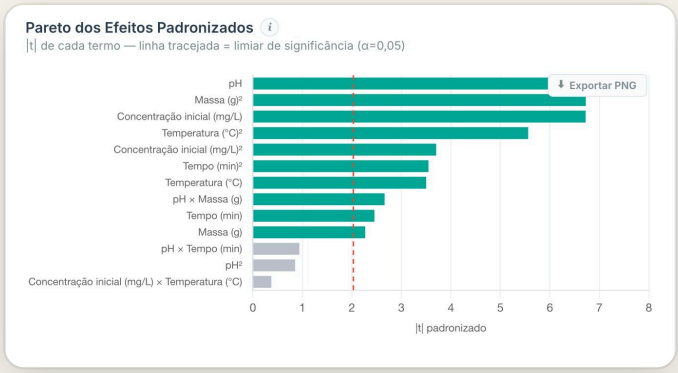
With 49 points, the test $R^2$ is treated as an exploratory estimate — and the dashboard says so up front.




**Parity and SHAP.** Left, predicted × observed for training and test sets; right, the SHAP beeswarm showing each variable's impact on the prediction.



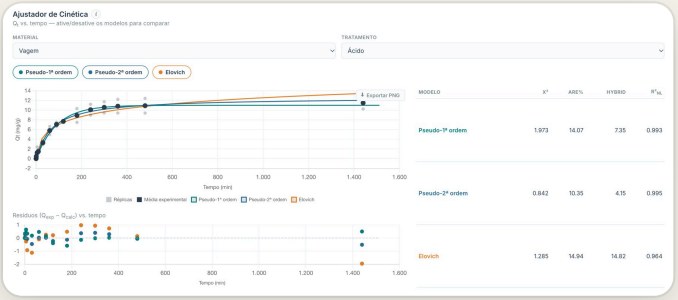

**Response surface.** Quadratic model with terms selected by VIF; $R^2$ = 0.959.



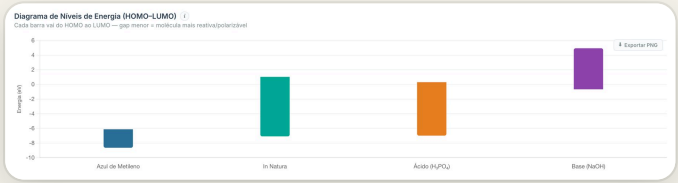

**Pareto of effects.** Type II ANOVA (statsmodels), α = 0.05.




**Equilibrium isotherms.** Langmuir, Freundlich, Temkin and Redlich-Peterson fitted by nonlinear regression, with residuals and $\chi^2$, ARE, HYBRID and $R^2_{NL}$ error functions.




**Kinetics.** Pseudo-second order with $R^2_{NL}$ = 0.995 and residual analysis.




**DFT B3LYP/6-31G*** (pyscf + RDKit, structures from PubChem). The HOMO of the alkoxide in the base treatment rises to −0.645 eV — an electronic explanation for the higher adsorption observed.

### RIGOR BEFORE $R^2$

- **OFAT is not DoE.** The data vary one factor at a time; RSM was applied retrospectively and the full 21-term model was singular. VIF-based selection kept the 13 terms with real support in the data.
- **Linearization misleads.** On the same data block, linearized Langmuir gave $R^2$ = 0.03; direct nonlinear fitting gave $R^2$ = 0.87. The problem was the method, not the data.
- **An optimum is a hypothesis.** The desirability optimum (differential evolution) is presented as a suggestion to be confirmed in the lab.

`scikit-learn` `XGBoost` `SHAP` `statsmodels` `SciPy` `Plotly` `Flask` `pyscf` `RDKit` `SQLite` `Google Drive API`

DATA ENGINEERING · PUBLIC SECTOR

# From shift-report PDFs to an incident map

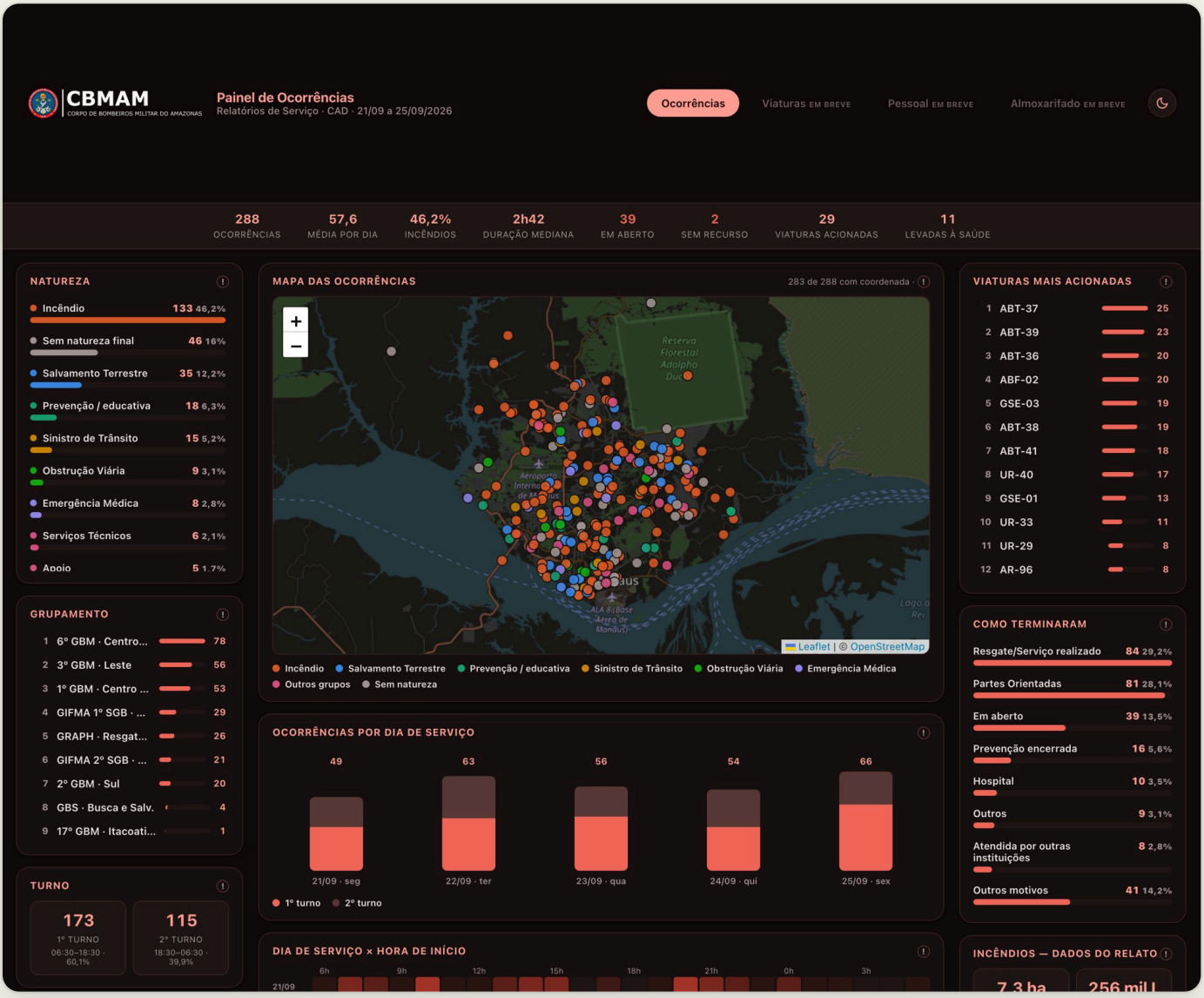
*Shift reports from the Amazonas State Fire Department became a dashboard with KPIs, a map and peak-hour patterns.*

**288**
unique incidents

**46%**
are fires

**2h42**
median duration

**29**
vehicles dispatched




**CBMAM incident dashboard.** Incident type, battalion and shift are clickable and filter everything; Leaflet map colored by type; incidents per service day and a day × hour heatmap.

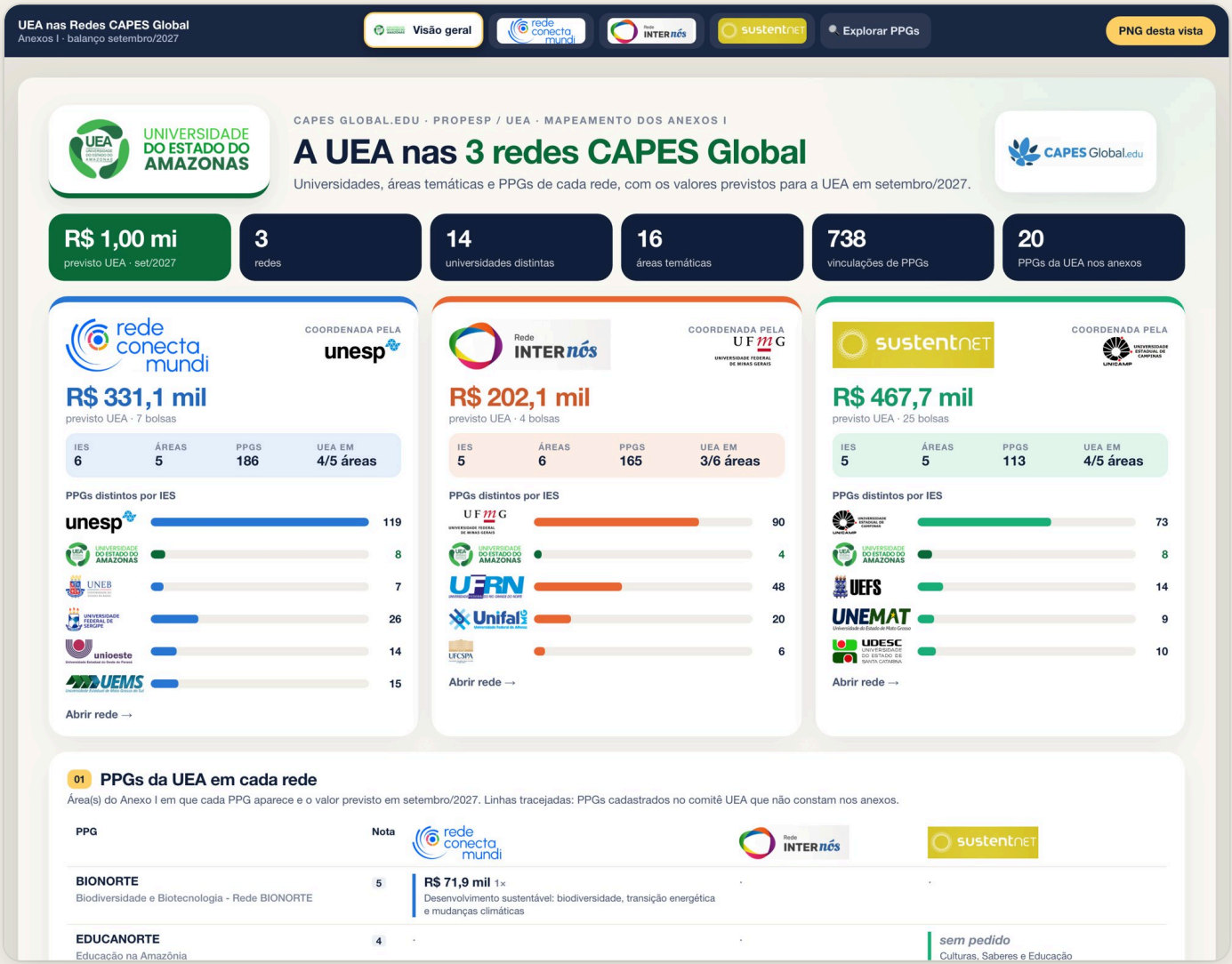
The source was PDFs exported from the dispatch system — one per shift, with summary tables and free-text blocks for each incident. The ETL uses **pdftotext** and regular expressions to extract status, address, coordinates, dispatched vehicles and outcome.

Open incidents reappear in the next shift's report; **deduplication by ID** prevents double counting. Burned area and liters of water have no dedicated field: they were extracted from the narrative text, with rules to discard implausible values.

The narratives contain victims' personal data. Hence **privacy by design** under Brazil's LGPD: PDFs never enter the repository and no personal data reaches the dashboard. The documentation also lists what the data *cannot* measure — such as response time — and what to request from the agency to measure it.

# Three networks, fourteen universities, one dashboard

*UEA's participation in the CAPES Global program — previously scattered across proposal annexes and spreadsheets — brought together in a single HTML file any manager can open in a browser.*




**Overview.** The Conecta Mundi (UNESP), INTER-NÓS (UFMG) and SustentNET (UNICAMP) networks, with funding planned for UEA and programs per institution; each network has its own tab with a map and a thematic area × university matrix.

A Python script (**build_dashboard.py**) reads the macro-enabled workbook with **openpyxl**, extracts planned scholarship amounts from the budget PDF and cross-references everything with the committee registry in the research office's management system. The output is a single self-contained HTML file, with embedded logos and CSV export.

The "Explore programs" tab filters and sorts all 738 links from the annexes, and a committee × annex check automatically flags registry mismatches.

**738**
program × network links

**14**
universities

**16**
thematic areas

**20**
UEA graduate programs

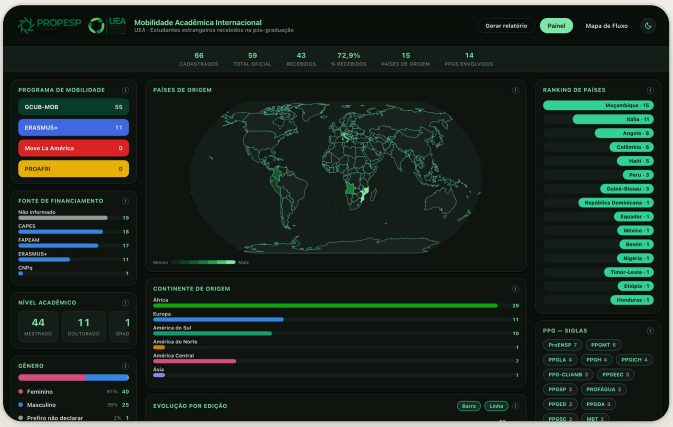
**3**
CAPES Global networks

**1**
HTML file, no server

`Python` `openpyxl` `pdftotext` `HTML/CSS/JS` `CSV export`

# Internationalization, dashboard by dashboard

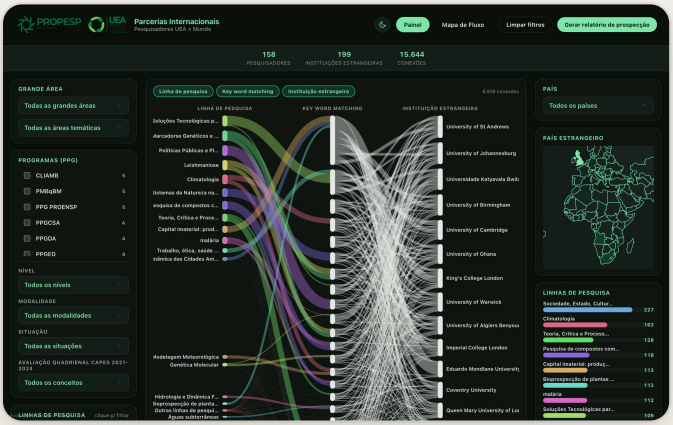
*One visual identity, one database and several audiences: managers, program coordinators and international students.*



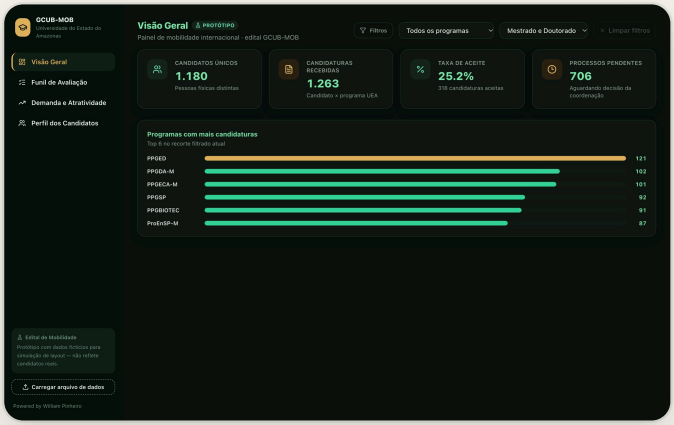

**Academic mobility.** International students hosted: 66 records, 15 countries, 14 programs. Data in a **star schema** (fact + dimensions) with a data dictionary and DAX measures.




**International agreements.** 42 agreements in 24 countries, a validity timeline and a flow globe from Manaus. Fed by a daily Google Drive → SQLite ETL (launchd).




**International partnerships.** First generation of the matching: 158 researchers, 199 institutions and 15,644 connections in a three-level Sankey.




**GCUB-MOB call.** Application review funnel in React + Recharts. The public version uses **synthetic data** with the real distribution; real data only loads locally.

## SHARED ARCHITECTURE

**SOURCES**
Spreadsheets on Google Drive (agreements, mobility, calls) and system exports

→

**DAILY ETL**
Python reads the sheets via API and consolidates fact and dimension tables

→

**SQLITE**
Single internationalization database, refreshed automatically once a day via launchd

→

**DASHBOARDS**
JSON per dashboard → static HTML/JS; PDF reports generated in the browser

### A single source of truth

Shared spreadsheets on Drive are read through the API and consolidated into a SQLite database refreshed automatically every day.

### Privacy by default

No national IDs, passports or e-mails on dashboards: anonymous keys, abbreviated names and synthetic data wherever it is public.

### Zero infrastructure cost

Static sites on GitHub Pages, with bundled libraries and PDF reports generated by the dashboard itself.

# Three generations of the same algorithm

*Partnership matching was not born finished: it was rewritten with every new management question — from one professor and one country to every program and every agreement.*

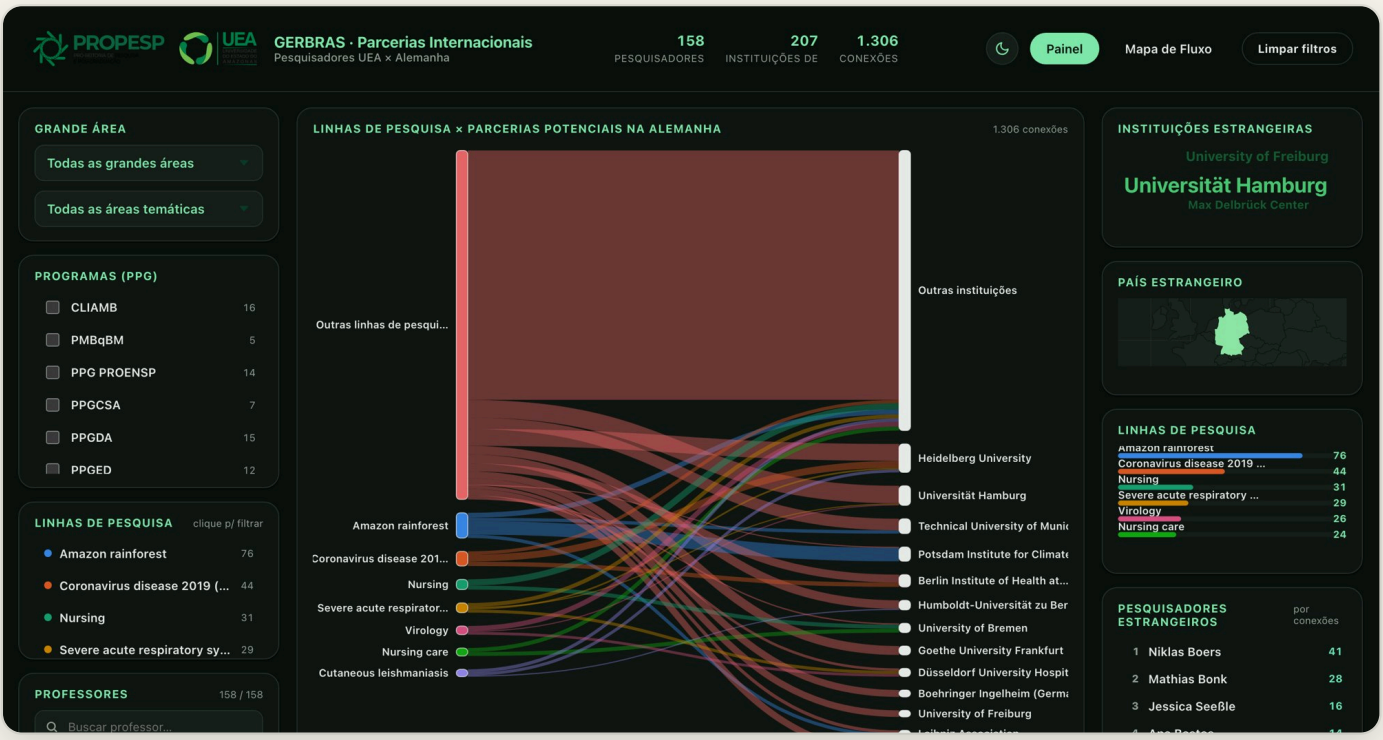
**V1 · GERBRAS**
Professor × foreign researcher via OpenAlex keywords. One country at a time (Germany).

→

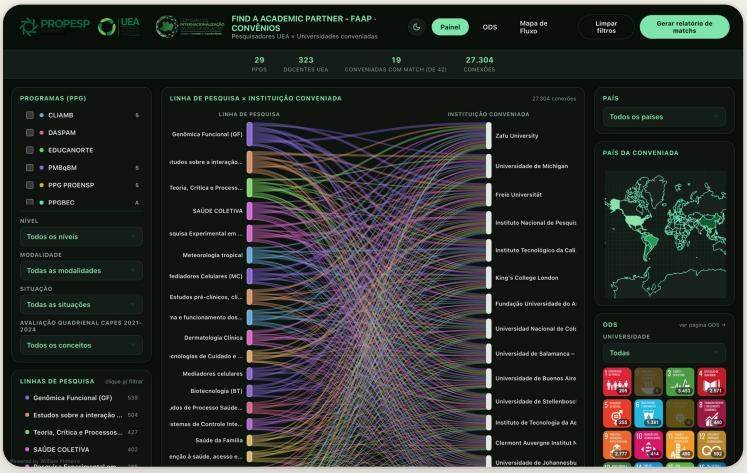
**V2 · FAAP**
Official research line × publications, with Sentence-BERT. 10 countries plus SDG classification.

→

**V3 · AGREEMENTS**
Same method applied to every institution with an active agreement, Brazilian ones included.




**GERBRAS — first version.** Sankey of research lines × German institutions: 158 researchers, 207 institutions and 1,306 connections, with foreign researchers ranked by number of links.




**FAAP Agreements.** 19 of the 42 partner institutions match UEA's research lines, totaling 27,304 connections. The indicator is compared directly with the agreements dashboard, showing where an agreement exists on paper but there is no shared research yet.

**1,306**
connections in V1 (1 country)

**58,084**
connections in V2 (10 countries)

**27,304**
connections with partners (V3)

**19/42**
partners with shared research

**PRODUCT LESSON**

The unit of analysis moved from "professor" to "official research line": results became stable, explainable and aligned with how the university is actually organized.

FROM DASHBOARD TO EXECUTIVE DOCUMENT

# When leadership asks for a PDF

*Interactive dashboards are for exploring; meetings with the rector and foreign partners call for a document. The same data becomes executive reports generated by code.*




**Cover.** United States report.




**Overview.** Summary by institution and program.

For each international mission, a script reads the numbers straight from the dashboard's data, builds the report in HTML, generates an editable Word version and exports the PDF with headless Chrome. No number is typed by hand.

In the US edition, programs from the UFMG cooperation network appear first for each institution, as requested by management — a business-rule change, not a layout change.

**13,445**
connections in the US report

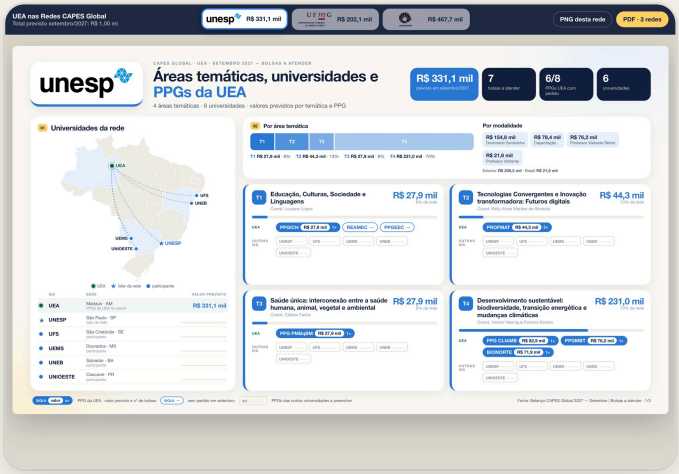
**8,239**
foreign researchers

**10**
US institutions

**3**
missions: US, Italy and UK



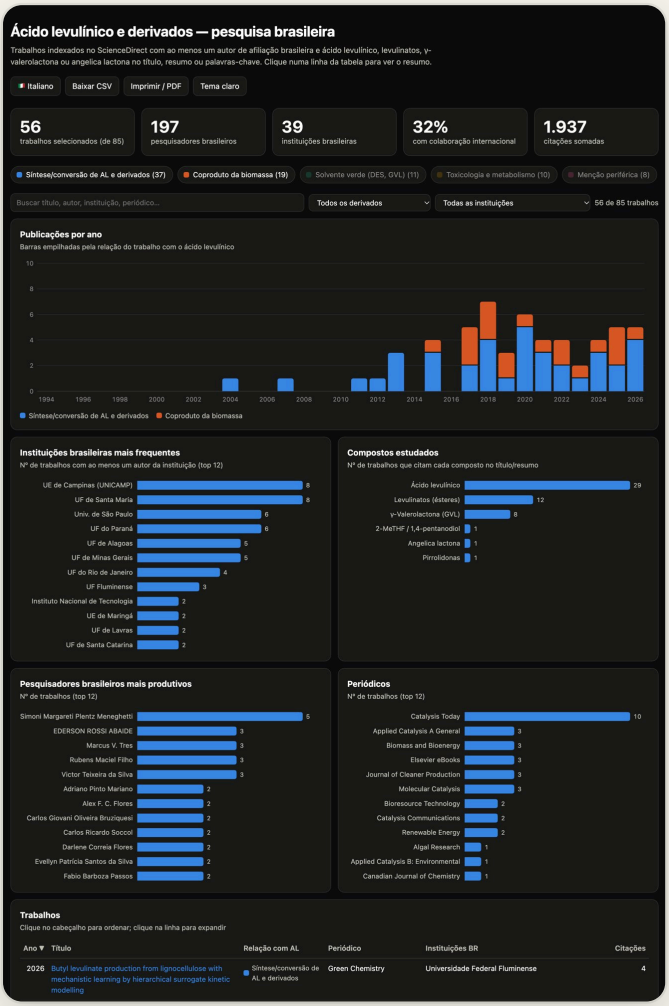

**Mobility flow map.** Students' origins and destinations, with line width proportional to volume.



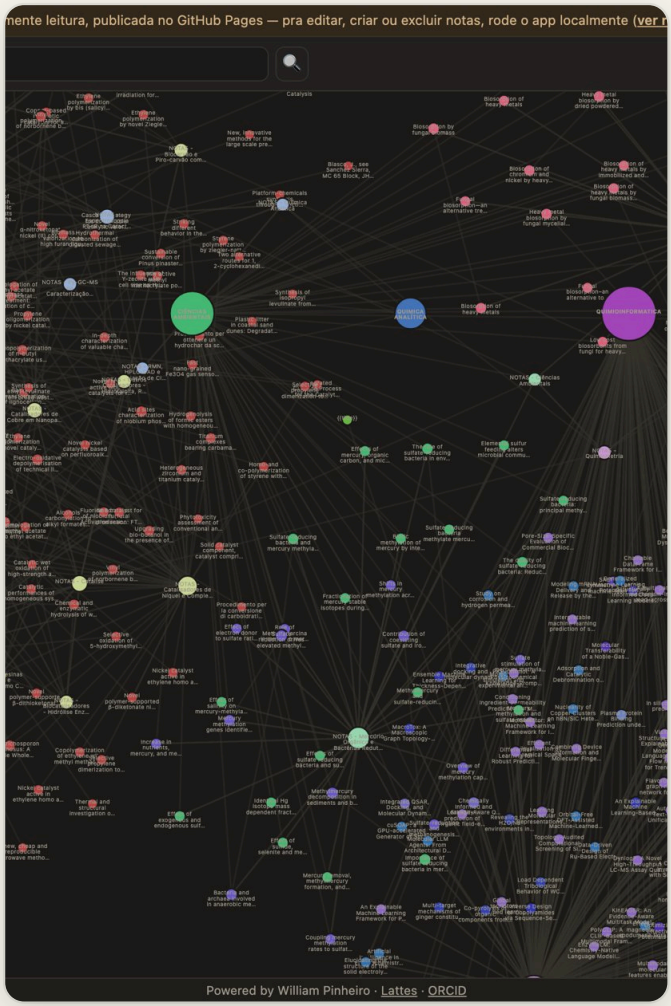

**CAPES Global network panel.** Meeting edition, with budget per thematic area and a map of partner universities — plus PDF export.

`python-docx` `HTML → PDF` `headless Chrome` `reproducible reports`

# Measuring research and organizing knowledge




**Bibliometrics: levulinic acid in Brazil.** Output per year, institutions, compounds, researchers and journals.




**Second Brain.** Interactive knowledge graph with hundreds of literature notes, colored by topic.

## Mapping Brazilian biorefinery research

DOIs collected through ScienceDirect advanced search were enriched with **OpenAlex** and **Semantic Scholar** metadata (authors, affiliations, citations, abstracts). Regex-based rules grouped papers by chemical derivative and synthesis route, and manual curation kept only the truly relevant ones.

**56**
curated papers (of 85)

**197**
researchers

**32%**
with international collaboration

**1,937**
total citations

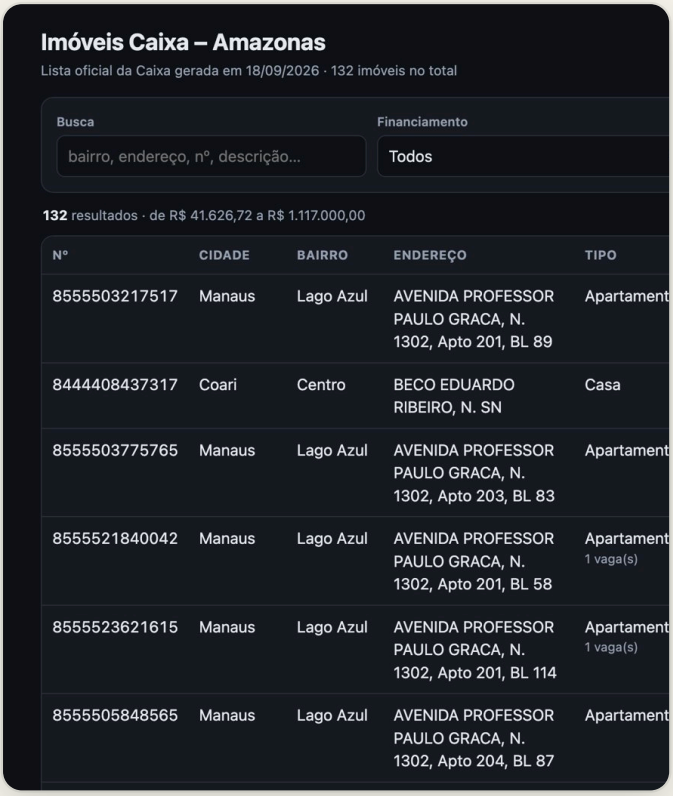
## A home-built “Obsidian”, from scratch

A knowledge-management web app: a dependency-free Python server, a REST API over Markdown files, a backlink index, full-text search and a **D3.js** force graph with topic and tag filters. Google Scholar alerts turn into reading notes automatically, and a read-only copy is published on the web.

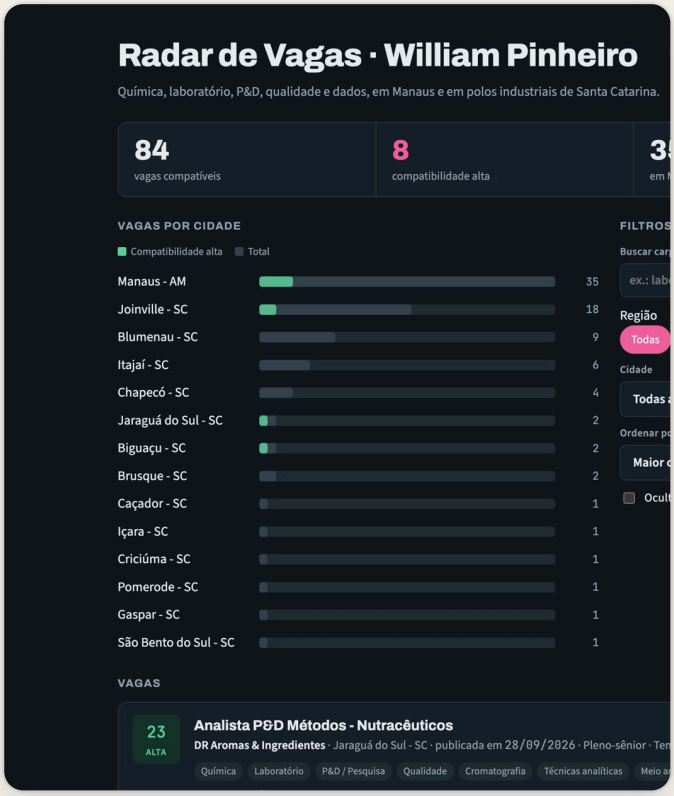
`OpenAlex` `Semantic Scholar` `regex` `pandas` `D3.js` `REST` `Markdown`

# Collect, filter, decide

*Outside institutions, the same toolbox solves everyday problems: scraping, scoring rules and an HTML page that fits on a phone.*




**Bank-owned properties in Amazonas.** Scraping of the official Caixa sales list, discount over appraised value and a filter for financing eligibility: 132 properties in a sortable table, with a mobile version.




**Job radar.** Collects public job postings across several cities and applies a weighted term score (title × description) that rates fit with the résumé as high, medium or low.

### Water Quality Index (WQI) system

First dashboard: water quality index per river basin and sampling point, with scikit-learn/XGBoost models, Prophet forecasting, Folium maps and PDF reports (Streamlit).

### UEA course catalog

Concurrent scraping of the university's course portal: 900+ undergraduate, specialization and graduate offerings, with caching and counts by level for internationalization demand.

### Funding-call monitor

Routine that searches international calls, checks them against what is already published, flags expired deadlines and generates ready-to-use blocks for the research office website.

### Research lines and Lattes

Canonical base of 166 research lines from 19 programs, linked to faculty Lattes and ORCID IDs — the matching's source of truth.

### Social media agent

Workflow that drafts the post, prepares the image and publishes to Instagram and Facebook via the Graph API, always with manual approval.

### Dashboards on the official site

Embedding of the dashboards in the research office's institutional pages, with corrected embed code.

ALSO IN THE PORTFOLIO

# Data inside real-world systems

PROPESP · UEA

### Scholarship and budget manager

Django + PostgreSQL rewrite of the scholarship planning system. Imports the official spreadsheet from Google Drive with a preview before writing and reconciles it row by row against the source (191/191). Dashboard with a **Sankey** of networks → themes → programs, a Gantt of validity periods and a reallocation simulator.

RETAIL

### Optical store management, from POS to finance

Django system with work orders, cash register, store credit, accounts payable and customer follow-up lists. Role-based access: costs and margins visible only to the owner, enforced by **automated tests**. Two-factor authentication and encrypted backups.

CBA

### Project reporting in Power BI

At the Amazon Bio-business Center, supported multi-sector project management with **semi-automated Power BI reports** and Python data flows, reducing manual consolidation work.

TOOLBOX

| | |
|---|---|
| **Languages** | Python · SQL · JavaScript · DAX · HTML/CSS |
| **Data & ETL** | pandas · SQLite · PostgreSQL · openpyxl · PDF extraction (pdftotext + regex) · Google Drive API · schema validation · scheduling (launchd/cron) |
| **APIs** | OpenAlex · ORCID · Semantic Scholar · PubChem PUG REST · Google Sheets/Drive · Meta Graph API |
| **Statistics & ML** | scikit-learn · XGBoost · SVR · Random Forest · GridSearchCV · SHAP · statsmodels (ANOVA, VIF) · SciPy (nonlinear regression, differential evolution) · RSM/DoE |
| **NLP** | Sentence-BERT (embeddings, cosine similarity) · spaCy (lemmatization) · semantic classification (SDGs) |
| **Visualization** | Power BI · D3.js · Plotly · Leaflet · Recharts · Sankey diagrams, flow globes, heatmaps · light/dark design |
| **Engineering** | Git/GitHub · GitHub Pages · Django · Flask · React + Vite · Docker · automated testing (pytest) · AI-assisted development (Claude Code) |
| **Scientific domain** | Chemometrics · DFT (pyscf, RDKit) · analytical techniques (HPLC, GC-MS, ICP-OES, XRD, FTIR) · catalysis and biorefining |
| **Governance** | Data privacy (LGPD) by default · documented KPIs and limitations · synthetic data for public releases |

# 35 projects, 17 dashboards

*Everything built in 2026, grouped by area. ● interactive dashboard · ◆ published online*

## INTERNATIONALIZATION · PROPESP/UEA

1 **FAAP — program × university prospecting ●◆**
Sentence-BERT, OpenAlex, SDGs · 58,084 connections

2 **FAAP Agreements ●**
matching with the 42 partner institutions

3 **GERBRAS — partnerships v1 ●◆**
professor × researcher by keyword

4 **International partnerships (general) ●◆**
7 countries, 15,644 connections

5 **International agreements ●◆**
42 agreements, flow globe, validity periods

6 **Academic mobility ●◆**
star schema, DAX, flow map

7 **GCUB-MOB ●◆**
React + Recharts, synthetic public data

8 **UEA in the CAPES Global networks ●◆**
Excel + PDF → self-contained HTML

9 **Network panel for meetings ●**
budget per thematic area, PDF export

10 **Executive prospecting reports**
US, Italy and UK · HTML → DOCX/PDF

11 **Google Drive → SQLite ETL**
service account, daily refresh (launchd)

12 **Research lines + Lattes/ORCID base**
166 lines, 19 programs

13 **UEA course catalog scraping**
900+ offerings, concurrent with cache

14 **International funding-call monitor**
duplicate and deadline control

15 **Dashboards embedded in the official site**
institutional page embeds

16 **Scholarship and budget manager ●**
Django + PostgreSQL, Sankey, Gantt

## SCIENCE AND RESEARCH

17 **Chemometrics (pod/palm fibers) ●◆**
ML/SHAP, RSM, isotherms, DFT

18 **Water Quality Index system ●**
Streamlit, XGBoost, Prophet, Folium

19 **Levulinic acid bibliometrics ●**
ScienceDirect + OpenAlex + Semantic Scholar

20 **Second Brain ●◆**
knowledge app with D3 graph

21 **Research vault + reading agent**
Scholar alerts → literature notes

22 **AI automation library**
versioned skills for recurring workflows

## PUBLIC SECTOR

23 **Fire Department incidents ●◆**
PDF ETL, Leaflet map, data privacy

## BUSINESS

24 **Optical store management system ●**
Django, POS, finance, lens catalog

25 **Social media posting agent**
Graph API, per-post approval

26 **Google Business Profile audit**
local digital presence analysis

27 **Digital product launch guide**
market research and web sources

28 **Power BI reporting · CBA**
multi-sector project management

## INDEPENDENT

29 **Bank-owned properties in Amazonas ●**
scraping, discount, financing filter

30 **Job radar ●**
collection + fit score

31 **Digital footprint audit**
OSINT on one's own online presence

32 **Documentary genealogy research**
historical records, evidence grading

33 **Commemorative brochure (HTML → PDF)**
editorial layout by code

34 **Generative AI explorations**
3D modeling and model reverse engineering

35 **This portfolio**
HTML → PDF, PT and EN

BACK COVER

# Shall we talk about your data?

*Point your camera and explore the live dashboards — the same systems featured in this issue.*

**FAAP · AI matching**
International partnership prospecting

**Chemometrics & ML**
ML/XAI, RSM, isotherms and DFT

**CBMAM incidents**
PDF ETL and geospatial dashboard

**CAPES Global**
UEA in the 3 networks

**GitHub**
github.com/willpine1992 — code and other dashboards

**Lattes CV**
Brazil's official academic CV: education, publications, projects

## EDUCATION

**PhD in Chemistry — Materials and Interfaces** — Federal University of Amazonas (UFAM), 2021–2026, with a CAPES sandwich doctorate at the University of Pisa, Italy
**MSc in Biotechnology and Amazonian Natural Resources** — Amazonas State University (UEA), 2019–2021
**BSc in Chemistry** — UEA, 2015–2018
**Chemistry Technician** — IFAM · **Associate degree in Oil and Gas** — UNINORTE

## PUBLICATIONS

7 journal articles (Bioresource Technology, Renewable Energy, Environmental Science and Pollution Research, among others) · 2 book chapters (Elsevier, Wiley) · 184 citations on Google Scholar

## CONTACT


**William Pinheiro**

CHEMIST · AI-ASSISTED DATA ANALYSIS

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## LANGUAGES

Portuguese (native) · English B1 · Italian A2 · French A2

INSERT · RÉSUMÉ

# William Pinheiro

## PhD in Materials Chemistry · AI-assisted Data Analysis

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### SUMMARY

Chemist with a PhD in Materials and Interfaces Chemistry (UFAM, with a sandwich period at the University of Pisa), experienced in heterogeneous catalysis, biomass and analytical techniques. My specialty is chemistry; I do data analysis using generative AI tools to help write code — I define the questions, choose the methods, validate the results and provide the chemical interpretation. I apply this to chemometrics and explainable machine learning, and to data dashboards for university management (17 dashboards built, 10 published online).

### OBJECTIVE

To work in **data analysis applied to chemistry** — R&D, laboratories, quality control, materials, bioeconomy and the chemical industry — combining domain expertise with statistical modeling, machine learning and visualization.

### SKILLS

| | |
|---|---|
| **Chemistry (specialty)** | Materials and heterogeneous catalysis, biomass and biorefining, adsorption; XRD, XRF, FTIR, SEM, TGA, HPLC, GC-MS, ICP-OES; DFT (pyscf) |
| **Chemometrics** | DoE/RSM, ANOVA, VIF, nonlinear regression (isotherms and kinetics), error functions, explainable ML (SHAP) |
| **Analytics & BI** | Power BI (star schema, DAX), KPI design, interactive dashboards, automated executive reporting |
| **Programming** | Python (pandas, NumPy, SciPy), SQL (SQLite, PostgreSQL), JavaScript (D3.js, Plotly, Leaflet), Git |
| **Data & ETL** | REST APIs, web scraping, regex-based PDF extraction, Google Drive/Sheets API, schema validation, scheduled jobs |
| **Machine learning** | scikit-learn, XGBoost, SVR, Random Forest, cross-validation, GridSearchCV |
| **NLP** | Sentence-BERT (embeddings, cosine similarity), spaCy, semantic classification |
| **Generative AI** | AI-assisted development (Claude Code) with critical review, testing and verification of results |

### EXPERIENCE

**Postdoctoral Researcher — Data & Internationalization** 2026 – present
*Office of Research and Graduate Studies (PROPESP) · Amazonas State University (UEA)*

- Built FAAP, a partnership-matching system using Sentence-BERT and OpenAlex: 29 graduate programs, 323 faculty, 384 foreign institutions and 58,084 ranked connections.
- Built and maintains 10 public dashboards (agreements, mobility, funding calls, CAPES Global networks) on a SQLite database fed by a daily Google Drive ETL.
- Automated executive reports for international missions (US, Italy, UK): dashboard data → HTML → Word/PDF, with no manual data entry.
- Rewrote the scholarship planning system in Django + PostgreSQL, with imports reconciled row by row against the official spreadsheet (191/191).

**Research Fellow — Project Management & Reporting** 2025 – 2026
*Amazon Bio-business Center (CBA)*

- Developed semi-automated Power BI reports for multi-sector project management and Python data flows for consolidation.

**PhD Candidate in Materials Chemistry — Analytical Center** 2021 – 2025
*Federal University of Amazonas (UFAM) · sandwich doctorate at the University of Pisa, Italy*

- Developed sulfated kaolin and niobium-based heterogeneous catalysts to convert Amazonian biomass waste into value-added products.
- Characterized materials and quantified products by XRD, XRF, FTIR, SEM, TGA, HPLC and GC-MS, processing the data with chemometrics.
- Co-author of 7 international journal articles (184 citations) and 2 book chapters (Elsevier, Wiley).

**Sales Consultant — health insurance** 2021 – 2022

- Managed sales funnel and lead portfolio in a CRM; lead generation through paid social media.

**Petrochemical Process Operator** 2012 – 2015
*Videolar S/A · polystyrene production*

## SELECTED PROJECTS

### Incident Dashboard — Amazonas State Fire Department

Python · regex · Leaflet

- ETL of shift-report PDFs with ID-based deduplication and metric extraction from free text; 288 incidents, documented KPIs and no personal data exposed.

### Bioadsorption chemometrics with explainable ML

scikit-learn · SHAP · statsmodels

- ETL with lab-spreadsheet validation, RF/XGBoost/SVR comparison (test $R^2$ 0.859), ANOVA and RSM with VIF-based selection, nonlinear isotherm fitting and DFT calculations.

### UEA in the CAPES Global networks

openpyxl · pdftotext

- Consolidated a macro-enabled workbook and a budget PDF into a self-contained HTML dashboard: 738 links, 14 universities and automatic mismatch checks.

### Bibliometrics and scraping

OpenAlex · Semantic Scholar

- Map of Brazilian levulinic acid research (56 curated papers, 1,937 citations); scraping of 900+ university courses; bank-owned property monitor with financing filter.

## EDUCATION

### PhD in Chemistry — Materials and Interfaces

2021 – 2026

*Federal University of Amazonas (UFAM) · CAPES sandwich doctorate at the University of Pisa (2024–2025) · thesis: sulfated kaolin and niobium heterogeneous catalysts for waste valorization*

### MSc in Biotechnology and Amazonian Natural Resources

2019 – 2021

*Amazonas State University (UEA)*

### BSc in Chemistry

2015 – 2018

*Amazonas State University (UEA)*

### Chemistry Technician · Associate degree in Oil and Gas

2013 – 2015 · 2010 – 2012

*IFAM · UNINORTE*

## LANGUAGES

Portuguese (native) · English B1 · Italian A2 · French A2

## STRENGTHS

- A chemist first: understands where lab data come from, their errors and what they mean chemically.
- End-to-end delivery: from raw source to published dashboard, with documentation and reproducibility.
- Transparent about AI use: code is written with generative AI assistance; methods, validation and interpretation are my own responsibility.

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# Scientific output and track record

**7** journal articles

**2** book chapters (Elsevier, Wiley)

**184** Google Scholar citations (11 works, Apr 2026)

**4** completed research projects

**3** data systems registered in Lattes

## PEER-REVIEWED JOURNAL ARTICLES

PINHEIRO, W.; FOGASSA, A. S.; SOUZA, M. P.; PAES, O. A. R. L.; DUVOISIN JR., S.; FREITAS, F. A. Produção de açúcares a partir de resíduos lignocelulósicos da Amazônia: coroa e folhas de abacaxi, casca de tucumã e sementes de açaí. *Cuadernos de Educación y Desarrollo*, v. 18, e10601, 2026.

PINHEIRO, W.; FOGASSA, A. S.; SOUZA, M. P.; PAES, O. A. R. L.; DUVOISIN JR., S.; FREITAS, F. A. Valorização de resíduos lignocelulósicos amazônicos: otimização do pré-tratamento para a máxima liberação de monossacarídeos essenciais. *Observatorio de la Economía Latinoamericana*, v. 23, 2025.

LOBO, W. V.; PAES, O. A. R. L.; SOUSA, A. S.; PINHEIRO, W. et al. Application of sulfonated passion fruit seeds as a heterogeneous catalyst in the esterification of oleic and levulinic acids: optimization of reaction parameters. *Renewable Energy*, v. 241, 122355, 2025.

FREITAS, F. A.; PAES, O. A. R. L.; LOBO, W. V.; SOUSA, A. S.; [...] PINHEIRO, W. et al. Application of chemically modified palm (*Elaeis guineensis*) bunch fibers in the biosorption of methylene blue: kinetic and thermodynamic parameters. *Biofuels, Bioproducts & Biorefining*, v. 19, p. 2333–2353, 2025.

LOBO, W. V.; PAES, O. A. R. L.; PINHEIRO, W. et al. Application of chemically modified waste tucumã (*Astrocaryum aculeatum*) seeds in the biosorption of methylene blue: kinetic and thermodynamic parameters. *Environmental Science and Pollution Research*, 2024.

MOREIRA, A. F. S.; SOUZA, C. B.; PINHEIRO, W.; FREITAS, F. A.; LACHTER, E. R. Catalytic conversion of levulinic acid in higher added value products: sustainable routes for esters production. *Revista Virtual de Química*, v. 14, p. 380–392, 2022.

BARROS, S. S.; PESSOA JR., W. A. G.; SÁ, I. S. C.; [...] PINHEIRO, W. et al. Pineapple (*Ananas comosus*) leaves ash as a solid base catalyst for biodiesel synthesis. *Bioresource Technology*, v. 312, 123569, 2020.

## BOOK CHAPTERS

LOBO, W. V.; BARROS, S. S.; DUARTE, R. V. L.; PINHEIRO, W.; SOUZA, L. K. C.; FREITAS, F. A. Transesterification process of biomass. Amsterdam: *Elsevier*, 2023, p. 1–12.

PINHEIRO, W.; SILVA, E. P.; FREITAS, F. A. Circular economy and the development of sustainable products through the application of e-waste. In: *Electronic Waste Management*. Wiley, 2023, p. 124–133.

## RESEARCH PROJECTS

**Ionomic analysis of *Vismia*** 2019 – 2021
*MSc · ICP-AES, seasonal metal-nutrient profiles in 3 species · FAPEAM*

**Cupuaçu epicarp** 2018 · lead
*Chemical characterization and biological activity of the methanolic extract*

**BTEX on a university campus** 2017 – 2018
*Undergraduate research · quantification of volatile compounds · FAPEAM*

**Araçá-boi essential oil** 2016 – 2017 · lead
*Hydrodistillation optimization (5 variables) and GC-MS quantification · CNPq*

## TECHNICAL OUTPUT

- Software (Lattes, 2026): **FAAP — Find an Academic Partner**, **International Agreements** and **International Academic Mobility**.
- Standard operating procedure for cleaning firefighter proximity suits, Amazonas Military Fire Department (2023), presented at the 21st National Firefighters Seminar.

## ACADEMIC SERVICE

- **Grant proposal reviewer** — FAPEAM, 2022–2023.
- **Undergraduate thesis supervision** (Nutrition, 2020) — calorimetric evaluation of guaraná preparations.
- **Visiting PhD period** at the Università di Pisa with Prof. Anna Maria Raspolli Galletti; UFAM advisor: Prof. Flávio Augusto de Freitas.
- Current: visiting researcher at UEA; volunteer researcher at CBA.

## OUTREACH

- Coordinated 70% ethanol production for UEA's COVID-19 contingency plan (2020, 140 h).
- "Chemistry Show" (2018) and National Science and Technology Week (2017).
- 5 conference presentations and 1 extended abstract (AMAZONEQ, SBQ Norte).

## ADDITIONAL TRAINING

ICP atomic emission spectrometry (Shimadzu, 12 h) · Gas chromatography (8 h) · ISO 9001:2015, risk management (15 h)